

Indian Institute of Technology Tirupati

Department of Computer Science and Engineering

Assignment – Giant Step in Malware

Course Name : Computer Systems Security

Course Code : CS402M

Date : 19 August 2026

ASSIGNMENT DETAILS

Team Size : 2 Students per Team

Total Marks : 100

Submission : GitHub Repository Link + PDF
Report + Demo Video Link.

**Deadline: 25 August 2026. After this date, no
further extension will be provided.**

THE SCENARIO

The Department of Home Ministry has commissioned a proof-of-concept to understand how modern adversaries could weaponise biometric face recognition to deliver targeted payloads against critical-infrastructure personnel. You are the red team. Your task is to build a face-authentication application that behaves normally when a known, authorised person appears, but silently triggers a ransomware simulator when a designated target persona is recognised. Everything runs inside a dedicated VM snapshot.

The core challenge is that you have been given only one reference image from which to build a full training dataset. You must generate that dataset yourself using a single image generative model before you can train anything. The generated dataset must also include facial aging variations, representing realistic changes in appearance across different age ranges, to evaluate the robustness of the face-authentication system in identifying the target and deploying the payload despite age-related changes in facial appearance.

SANDBOX RULES

- **Dedicated VM Only.** All development and testing must run inside a dedicated VM with a snapshot taken before each run. Restore the snapshot after every test.
- **Targeted Professor** Each team is assigned respected Professor image as reference image for your assignment.
- **Controlled Environment.** Do not run any component of this assignment on a host machine, a shared lab computer, or any machine you do not personally own and control.
- **Repository Restriction.** Your GitHub repository must contain no functional encryption code. you can pull the payload on the fly and include all the details about code, dataset generation, model building, hyperparameter tuning. Also mention all the related reference paper and implementation details in the Readme.md file.

ASSIGNMENT OVERVIEW

PART	DESCRIPTION	MARKS
1	Synthetic Dataset Generation	25
2	Face Authentication App + Ransomware Simulator	30
3	NeurIPS-Format Technical Report	20
4	Live Demonstration (Video + Portal Upload)	10
5	Competitor Group Evaluation	5
6	Instructor Evaluation	10
TOTAL MARKS		100

PART 1 - Synthetic Dataset Generation [25 Marks]

You have one image. An adversary working against a real target rarely has more. Your first job is to turn that single image into a dataset large enough to train a reliable face classifier. Each team member contributes one reference image one of your own faces as Class A (the authenticated user), and one targeted Professor images as Class B.

REQUIREMENTS

01 Single-Image Generative Model [10 Marks]

Generate a minimum of **500 synthetic variations** covering different scales, lighting conditions, and minor pose offsets.

02 Conventional Augmentation [5 Marks]

Apply conventional augmentation: horizontal flip, brightness jitter, rotation up to $\pm 15^\circ$, Gaussian noise to build out both Class A and Class B to a minimum of **500 images each**, with an **80/10/10 train/val/test split**.

03 Dataset Publication and Reproducibility [5 Marks]

Publish the complete dataset (both classes, all splits) to a public GitHub repository. The README must include: the single input image used, the algorithm behind the model building and all hyperparameters, and the augmentation pipeline.

Tag the dataset commit with `v1.0-dataset`.

04 Report Discussion

[5 Marks]

Discuss in the report: what failure modes does training on synthetic data introduce, and what steps did you take to mitigate distributional shift at inference time?

PART 2 - Face Authentication App + Ransomware Simulator [30 MARKS]

Now using the data set and ML models available build the system that uses it a face-authentication app that makes a silent, irrevocable decision the moment it sees the target persona. Build a two-class face recognition model trained on the dataset from Part 1. Embed it inside an application with two clearly distinct authentication paths.

AUTHENTICATION PATHS

A AUTHENTICATION PATH A

Known User (Class A)

1. Camera opens, face is detected and matched to Class A.
2. App displays an “Authentication successful” screen and logs the event with a timestamp.

B AUTHENTICATION PATH B

Target Professor Case (Class B)

1. Camera detects a face matching Class B above your chosen confidence threshold (more that 0.85 is only considered valid).
2. App silently pulls the local payload and deploy it on the host machine on which its running.

PLATFORM REQUIREMENTS

Android App (Emulator)

[30 Marks]

- App installed in an Android emulator.
- Both paths of the authentication paths must be demonstrated.
- If you are not able to find Andriod based ransomware file write custom code to implement the encryption part that should exactly work as ransomware for Andriod.

- You need to demonstrate unavailability of android file system to the User in the simulator where your app deployed (once target is identified).

Linux or Windows Desktop App [20 Marks]

- Need to build a Windows or Linux App with proper UI and should be in complete working state (not just the executable).
- Both authentication paths must be demonstrated.
- You are free to pull the Ransomware payloads from git repo for respective OS that you are working on.

Note: You earn marks for one platform only. Choose the only one platform your team can deliver. Please don't include I tried multiple in the report.

PART 3 - NeurIPS-Format Technical Report [20 MARKS]

The system works. Now justify every decision you made architecture, hyperparameters, dataset, and accuracy in the format the research community uses.

Submit a report in the **NeurIPS 2026 formatting**. Use the official $\text{\LaTeX}$ template only; Any other template/PDF submission will not be considered for evaluation:

Formatting Instructions for NeurIPS 2026

REPORT REQUIREMENTS

- The report must follow the **NeurIPS 2026 formatting standard**.
- Use the **official NeurIPS 2026 $\text{\LaTeX}$ template**.
- Justify every major decision related to the **architecture, hyperparameters, dataset, and accuracy**.
- Report should contain proper NeurIPS formatting structure – Abstract, Introduction, Related works that you studied for implementing this assignment, Methodology that your team followed, results section should discuss how you evaluated your model 80/20 split, discussion corner cases and limitations of your work followed by conclusion. Appendix needs to be there with proper prompts if you use GenAI. Ethical considerations.
- Maximum **eight pages**, excluding references and the mandatory appendix.
- Need to share all your prompts etc in the appendix (Mandatory)

- Clear contributions of each team members for the assignment also need to be mentioned in the appendix (Mandatory).
- Copying the code or the contents from other student/teams will result in 0 marks for this assignment. All the report and code will be vetted with plagersim check.

PART 4 - Live Demonstration

[10 MARKS]

Record a continuous, unedited screen-capture video demonstrating both authentication paths in sequence. The video must show:

VIDEO SUBMISSION INFORMATION

- **Maximum Video Length:** 5 Minutes.
- Upload the video to the portal and paste the video link in your report appendix.
- **Video Speed:** Must not be more than **1.25×**.

DEMONSTRATION REQUIREMENTS

- **Clean Environment:** The app launching from a clean VM snapshot or a freshly started emulator.
- **Class A Authentication:** Team member appears on camera, face is recognised, and “Authentication successful” is displayed.
- **Class B Authentication:** The Class B reference image is held in front of the camera (printed or displayed on a second screen). Within a reasonable detection time, the simulator fires, the file manifest is written, and the ransom note UI appears full-screen.
- **Manifest Verification:** The terminal or log window showing the manifest file being created (`cat ransomware_manifest.txt`).
- **Brief Verbal Walkthrough:** One team member narrates while the other operates, explaining the architecture chosen, confidence threshold, and why Class B triggered.

PART 5 - Competitor Group Evaluation [5 MARKS]

Each team evaluates one other team’s submission and submits a short written peer review, with a maximum length of **one page**, covering:

PEER REVIEW REQUIREMENTS

1. **Dataset Quality:** Diversity of the generated images, size of each split, and any visible artefacts or distributional collapse.
2. **Model Performance:** Do the reported metrics seem consistent with the confusion matrix? Is the ablation study convincing?
3. **App Robustness:** Did the demo show edge cases such as occlusion, lighting change, or partial face? Did the threshold choice seem well-justified?
4. **Report Quality:** Clarity, completeness of hyperparameter documentation, and AI-prompt appendix.

PART 6 - Instructor Evaluation

[10 MARKS]

The instructor will evaluate the overall quality, completeness, and understanding demonstrated by the team throughout the assignment.